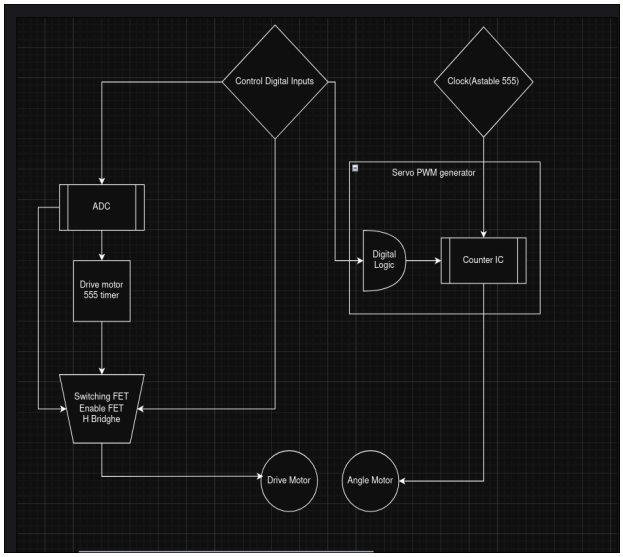
Daniel Salkin
06/03/2026

Analog Swerve Final Report

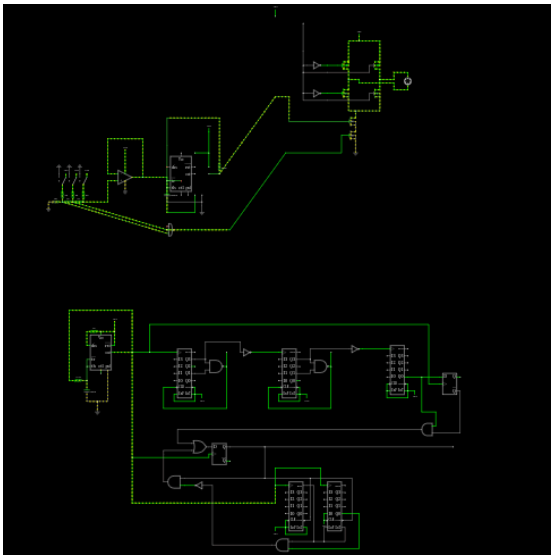
# Proposal Statement

**Abstract:** Swerve drives offer superior mobility by allowing simultaneous independent control of translation and rotation, but their high costs and complexity make them rare to find. My project creates a proof of concept of a low-cost swerve module using simple electronic components. The system is designed top-down: first with a controller that takes in inputs and breaks them down into simple digital signals. Those signals are sent into the two controllers: the angle and drive controllers. Each of those digests the digital signals and turns them into controls for the drive and angle motors. The drive motor is a simple DC motor from the 110 lab. To control it we use a pwm wave. An R-2R DAC drives the control input of  a 555 timer and controls the duty cycle of the PWM output which controls the drive motor. The angle motor is a servo motor that turns the module and is controlled by a variable width pwm wave every 20ms. This is achieved with a 10khz clock that is stepped down to 50hz to send a signal to start an output every 20ms. That also starts a separate counter which turns off the output after a variable amount of time demarcated by input signals. This lets us use our decoded signals to drive the module's angle control. These control circuits are hooked up to the motors which are mounted onto 3d printed components to form the entire swerve module. The results showed a feasible design, that while it needs improvement, could be used as a mobility system for a robot.

**Block diagram**




**Circuit diagram**



# Milestones

**Milestone 1:** Variable duty cycle generation for drive motor control **(Completed Week 5)**

This milestone will demonstrate the capability of this system to create the control signals which will determine the speed control of the drive motor. This will be validated by probing the output of the 555 timer with an oscilloscope and viewing varied duty cycles being produced with different inputs.

**Milestone 2:** Full drive motor control(Speed, direction, enable/disable) **(Completed Week 7)**

This milestone will demonstrate a fully completed drive motor control system. Variable speed, direction, and an enable disabled signal will fully function and work as expected when attached to a motor. This will be verified by first probing with an O-scope as I did for milestone 1, and then by hooking up a motor to the control circuits and controlling it.

**Milestone 3:** Variable width servo PWM control **(Completed Week 9)**

This Milestone will demonstrate the success of the servo controller. Variable width pulses sent every 20 milliseconds will control the angle at which the servo is pointing which will allow for angle control of the module. Input will be controlled by a series of binary signals which are inputs to the counting ICs as seen in the servo control circuit diagram. This will first be verified by reading the outputs on an oscilloscope and verifying the pulse width and pulse frequency, and then attached to the servo motor to test fully

**Milestone 4:** Full servo and drive motor integration onto swerve module **(Completed Week 9)**

This milestone will demonstrate the full functionality of the system with the angle and drive motors attached together into one coherent module. Manufactured and bonding will be robust enough to stand up to repeated demonstrations under load. This will be tested by demonstrating all of the states of the module and ensuring there is no damage to the components, electronics, or 3d printed parts.

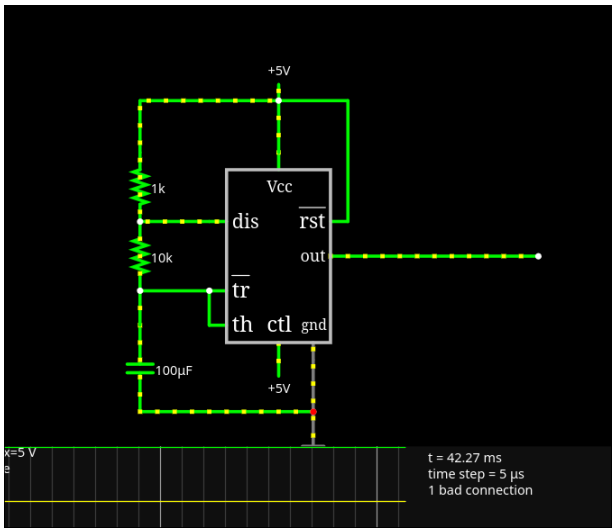
**Milestone 5:** Full module integration with controls **(Completed Week 10)**

This module will demonstrate the final fully successful state of the swerve module. Not only is the physical operation validated as per Milestone 5, but there is proper interface hardware such as rocker switches, encoders, or rotaries so that it is convenient for an outside tester to work with the hardware.

# Timeline

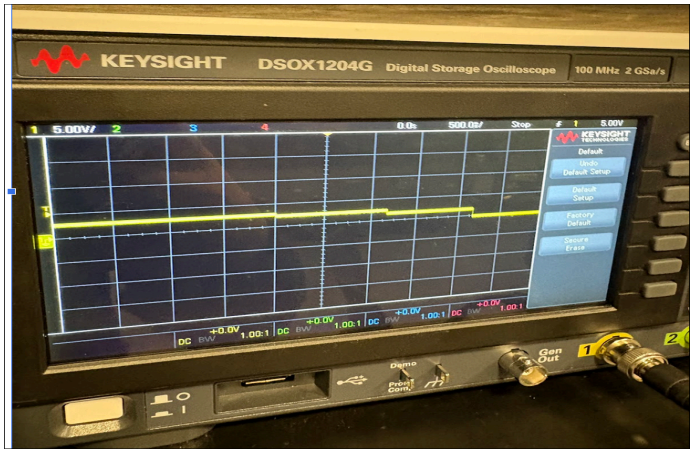
**Week 1(2/18):** Ideation of project really started. I designed a simulation of my 555 timer clock circuit in Falstad to figure out resistor and capacitor sizing, and how to start hooking up the circuit (1). After discussing with TAs, an R-2R ladder was chosen as the method to convert digital signals into something to be used, and component values were selected to make a reasonably high duty cycle for pwm switching to avoid visible switching/audible noise by the motors.

**Fig 1:** 555 timer oscillator




**Week 2(2/25):** I explored the difference between a R-2R DAC and a R-2NR DAC and decided that R-2R is a better solution. Although it is a bit less theoretically simple, hinging on thevenin equivalents to get the proper working, it requires less components and space to work properly. I manufactured and tested the DAC and op amp buffer and got that working. To validate the functionality, I varied the inputs and viewed the change in voltage on an oscilloscope (2). To create the 5v logic power rail, i used a resistor divider over grabbing an LDO from the component store due to a misunderstanding of what professor Schmitz recommended. This would prove to be a mistake and would lead to many delays down the road

**Fig 2:** DAC Voltage outputs



**Week 3(3/05):** I hooked up the 555 timer to the DAC and created a variable pwm pulse and validated it on the oscillosope(3,4). When using it to drive a motor I found out that the signal was demolished (5). After testing i concluded it was due to the draw on the 5v rail and started to work out what solutions to this i could come up with. After talking with professor Schmitz, I decided to research a zener diode linear regulator and to work on that the next week.

**Fig 3:** high duty cycle pwm

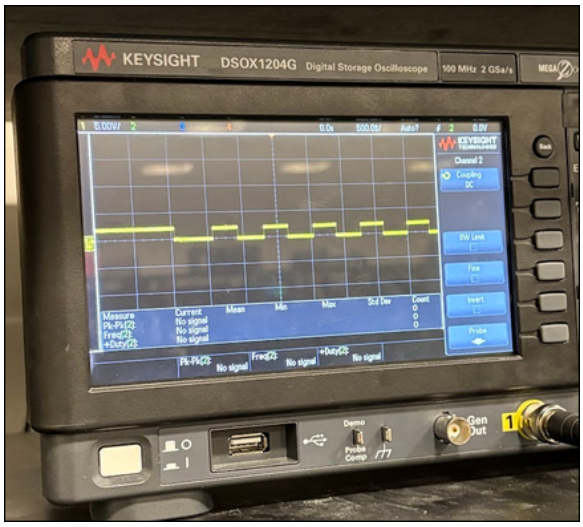
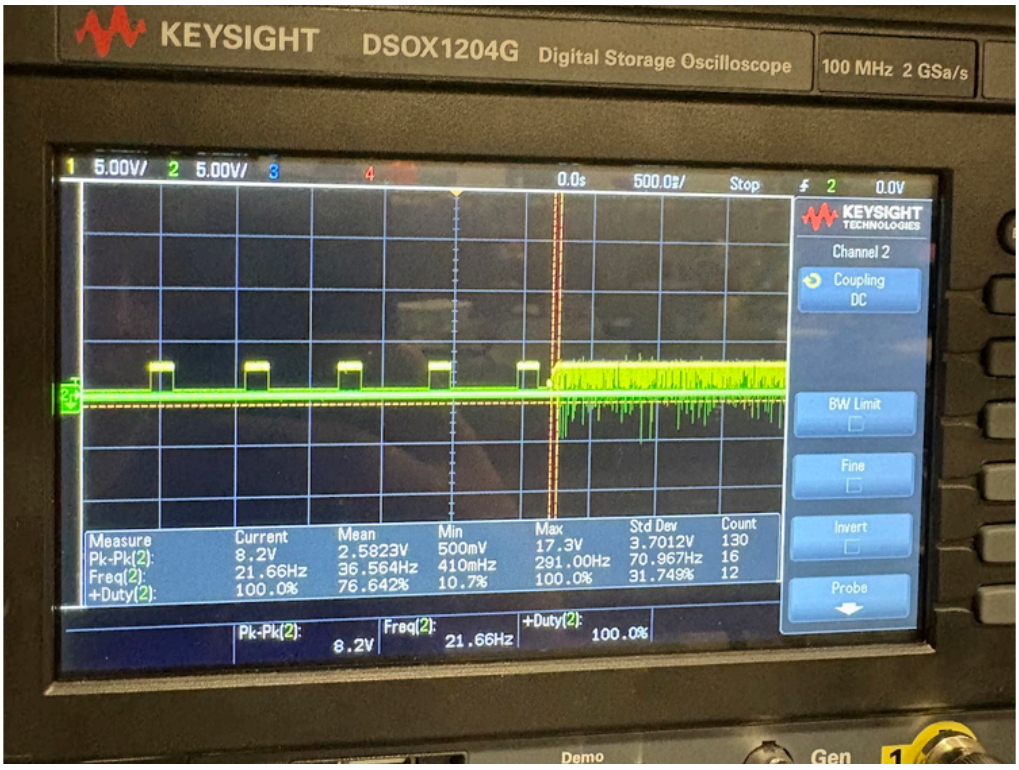



**Fig 4:** low duty cycle PWM




**Fig 5:** PWM pulses before and after driving a motor



**Week 4(3/13):** After doing some research on a zener diode linear regulator, I found the proper resistor values to create a 5v output and tested that. Nominally the 5v rail worked, and i was able to get it to control a motor but not well. After adding a nand gate to act as an enable bit, the current draw was too high and the 5v rail dropped down to 2v. Even with a separate power rail for the nand gate, the drop was high enough that there were negative effects. Since none of my solutions were working, the next logical step was to get an LDO IC and have that handle the stepdown.

**Week 5(3/27):** I grabbed a LDO from the supply room and tested it, but the 5v output was giving me a 6.2v output when measured which would drop a significant amount under load. After discussing with one of the professors, I moved to a different LDO and we believe that the issue was just a faulty IC. When testing with an O-scope I saw the proper waveforms being created, and was able to see the expected behavior on the geared-down 110 motors. Simultaneously, I started design work on the servo control segment of my project. I started sketching out the clocking and state machine and started thinking about component selection for my counters(6). Searching in the TI LMC555 timer datasheet I found a schematic for a 50% duty cycle square wave generator which is exactly what I wanted so I went with that, and looking through the parts list, i found that TI SN74ALS163 counters would work for what I needed (7). Following that equation, a possible capacitor and resistor i could have used was is a .1uF capacitor and a 720 ohm resistor.

**Fig 6:** Servo State Machine Diagram

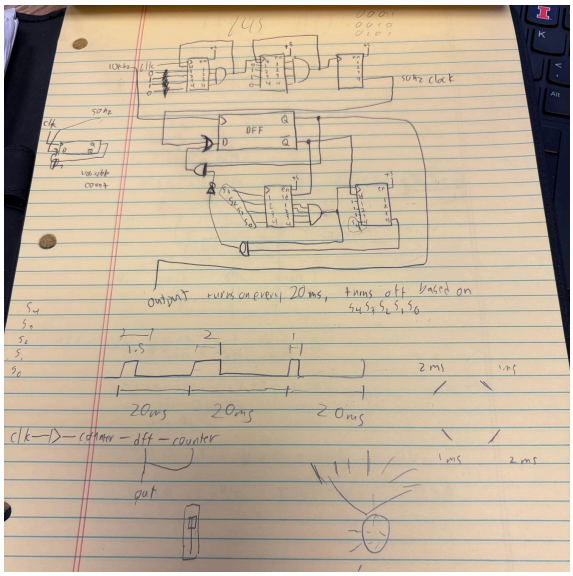
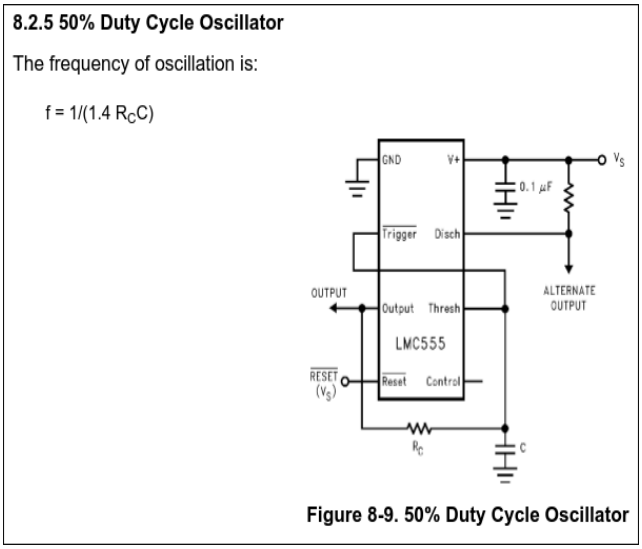



**Fig 7:** 555 Timer astable oscillator(50% duty cycle)



**Week 6(4/3):** I spent the weekend debugging the control circuits and getting the enable bits and h bridge to work properly. I finished wiring the h bridge configuration, but accidentally plugged in the motor to the wrong parts which ended up shorting and frying the h bridge (8,9). Over the week, I moved my diagram into a falstad simulation to verify the servo control output, and the simulation looked like it would work (10). In the lab, I replaced the H bridge and started verifying each circuit component to make sure that nothing was fried when I shorted the H bridge. When I added a flyback diode to the motor, I ended up causing another short and blowing up the diode because i did not think about how an h bridge would affect a flyback diode. Since the current can switch direction due to the h bridge, in one configuration it will act like a short circuit and cause issues.

**Fig 8:** PWM speed control (minus H-bridge)

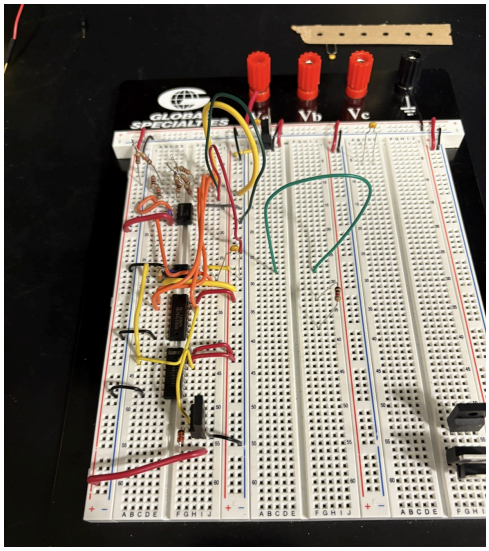



**Fig 9:** Blown up Component

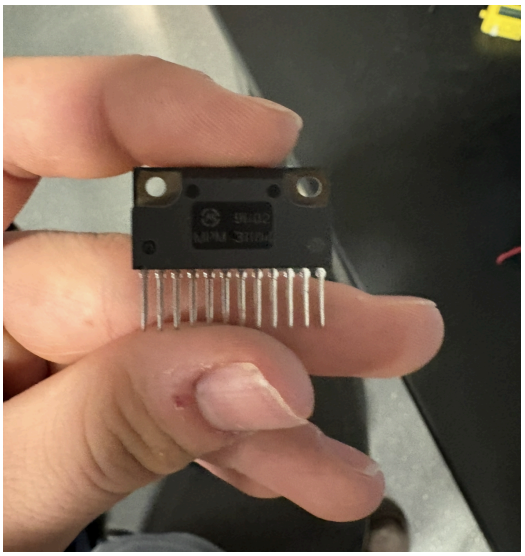
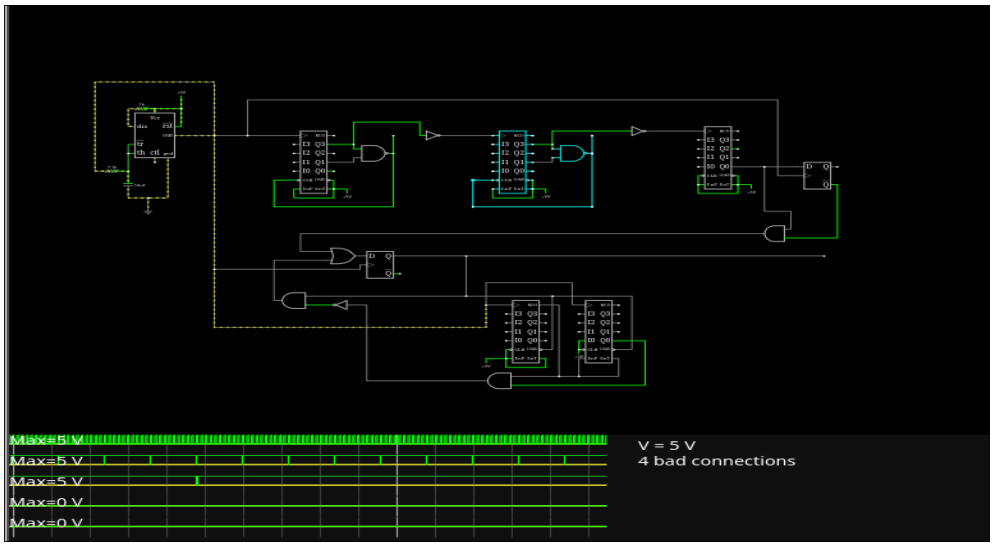


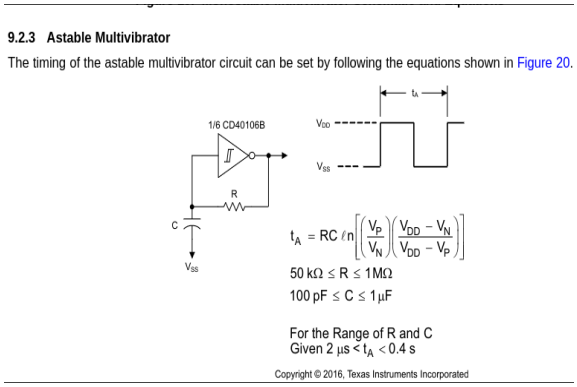
**Fig 10:** Servo Controller Circuit diagram



**Week 7(4/10):** This week I first completed the control circuits that were being worked on the week before. Although the H-bridge was blown up due to faulty wiring, everything on the board was wired correctly which led me to believe that I had placed it in the wrong spot on my breadboard as that was the last component to go in. Once the IC was replaced the drive motor controller worked as expected.

I was then recommended to use a Schmitt trigger inverter to create the clock signal for my servo circuits by a TA which I switched to because it was a simpler circuit and allowed me to tune the frequency more easily (12). I calculated the proper resistor and capacitor values based off of the datasheet(1nf and 120ohm). And then I used a potentiometer to tune the frequency ~10kHz, validating it by using the measure section of the )-scope to view the frequency of the square wave(13).

**Fig 12:** Application information



9.2.3 Astable Multivibrator

The timing of the astable multivibrator circuit can be set by following the equations shown in Figure 20.



$$t_A = RC\,\ell n\left[\left(\frac{V_P}{V_N}\right)\left(\frac{V_{DD}-V_N}{V_{DD}-V_P}\right)\right]$$

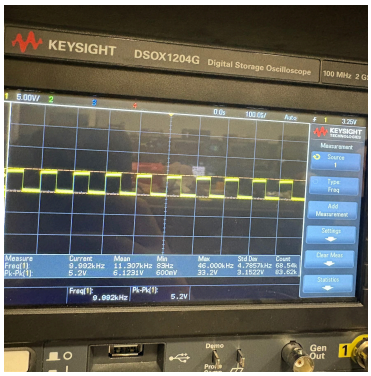
$50\ k\Omega \le R \le 1M\Omega$

$100\ pF \le C \le 1\ \mu F$

For the Range of R and C
Given $2\ \mu s < t_A < 0.4\ s$

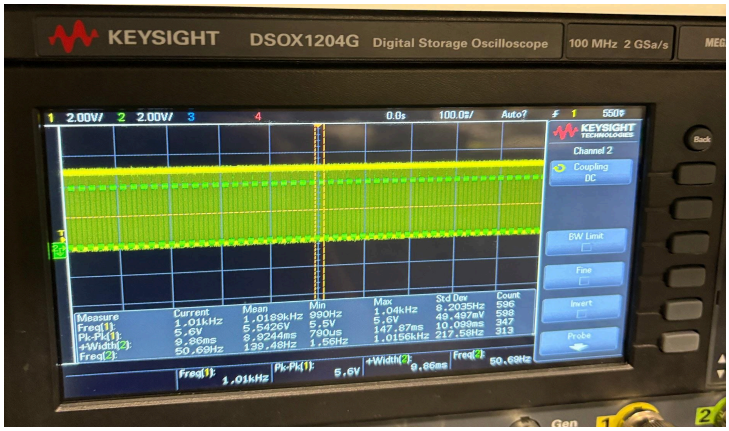
Copyright © 2016, Texas Instruments Incorporated

**Fig 13:** 10khz square wave




**Week 8(4/17):** I first spent some time tuning the frequency to be as close to 10khz as i could possibly get it, and then started implementing the counters to step down the frequency to 50hz. Due to time constraints and struggles with the oscilloscope I was only able to get down to 100hz, but I demonstrated the success of the step-down concept and learned more about the Oscilloscopes (14). When i was running two inputs, the autotrigger was getting confused, and i could get a much clearer reading by setting the trigger to normal mode and triggering off either my 10kHz or 100Hz wave to read the frequencies that I needed.

**Fig 14:** 1kHz and 100Hz square waves read on oscilloscope



**Week 9(4/23):** Over the week I finished up the servo control circuit but could not get the proper reading off the oscilloscope of the output. I worked on a test plan and narrowed down the issues to a few different things, but when testing the problem was much simpler than I expected. I found that the circuit was working as intended but the inputs were floating which read as high, so it would only count for 1 tick before overflowing. Although that worked in Falstad, the floating voltage was so high on my board that the counting chip I was using read that as high. Adding pulldown resistors solved that issue and servo control started working (17,18). In parallel, I had CADed the entire swerve module and had professor Schmitz print it so that I could work on Assembly and Validation over the weekend (15). Since this is a proof of concept, tolerances, strength, and manufacturability were not optimized since I was just attempting to get a working product.

**Fig 15:** CAD of the swerve module

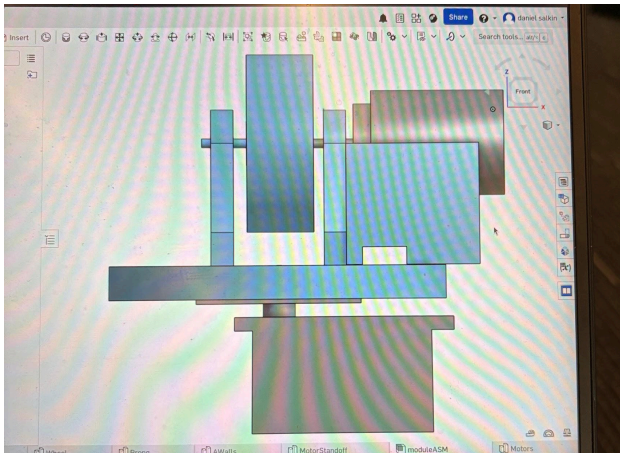


**Fig 16:** Sketches of the swerve module

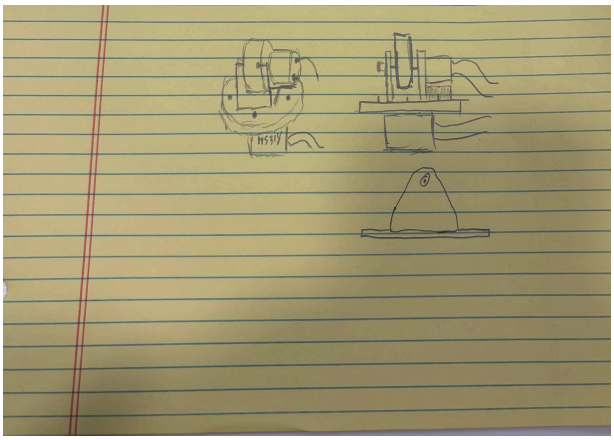


**Fig 17:** Minimum pwm with

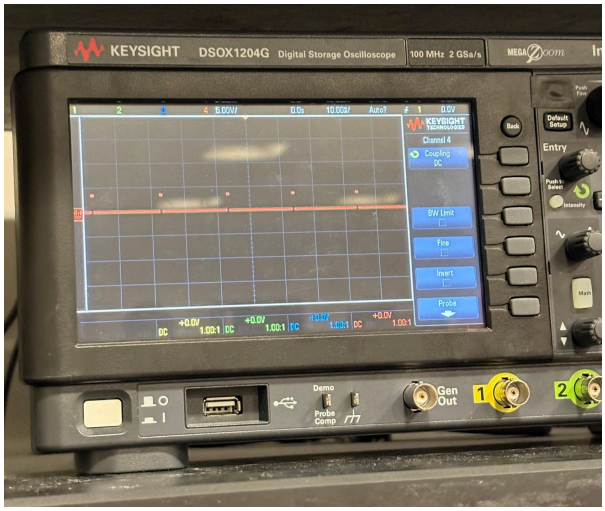
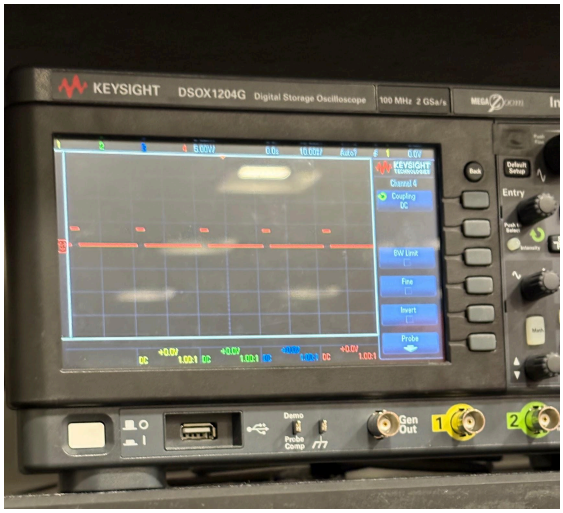



**Fig 18:** Greater than maximum PWM width



**Week 10(4/30):** Over the weekend, I worked on implementing a rotary control for the angle. I originally planned to use a rotary switch and some sort of combinatorial logic to control the inputs to the counting ICS that drive the angle motor, but when searching the supply cabinets, there were none. I then pivoted to a potentiometer that I found because it looked nice and comfortable and decided to create an ADC to determine the input bits from the potentiometer. To do so, I did some research and settled on building a tracking ADC (19). I created a 5-bit R-2R ladder DAC using the same process as for my 555 timer, and then hooked it up to my up and down counter and used an op amp as a comparator. This led to issues which I ultimately found was due to the fact that the .8V low output of the opamp was high enough to be read as a high by the up/down counter. Once I switched the opamp for an LM311 comparator, the issue disappeared and my tracking ADC worked as intended (20). I used this to read the voltage off of my potentiometer as an angle controller, but I needed to somehow bound those values, so i put my control potentiometer in series between two 110 pots and used those to tune the high and low values above and below the min and max values for the servo inputs so that i wouldnt overdrive the servo.

**Fig 19:** Tracking ADC schematic

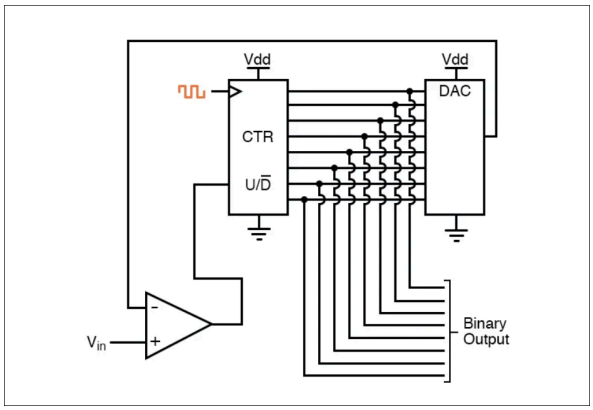
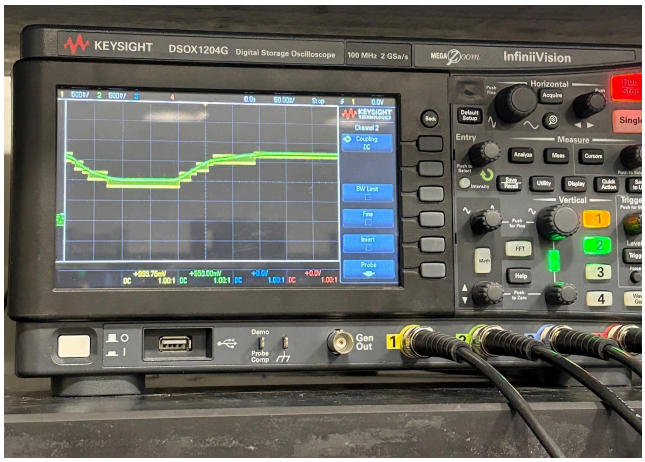


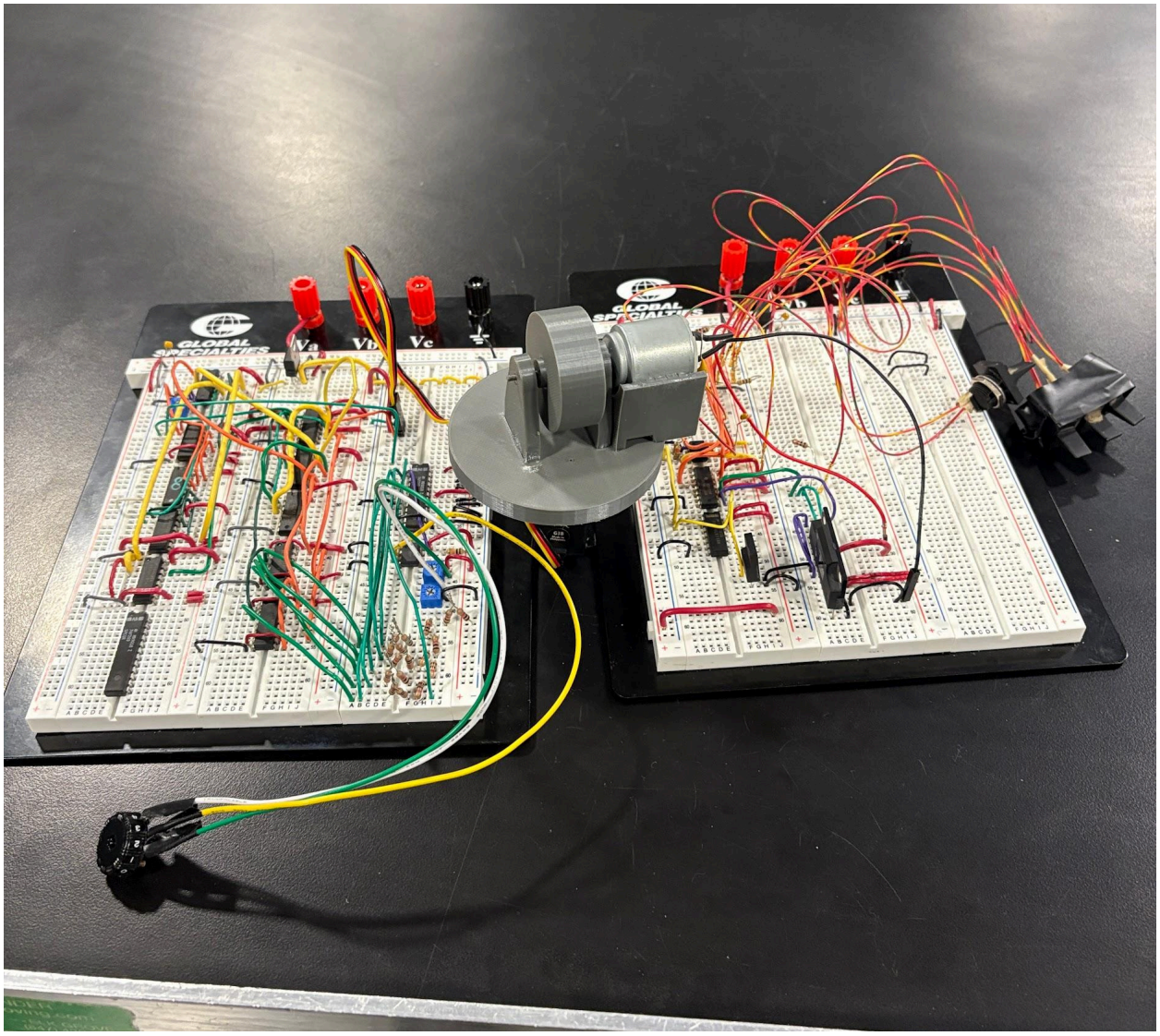

**Fig 20:** Tracking ADC(yellow) Analog Input(green)



# Final Product

Pictured below is the final state of the analog swerve project (21). The swerve module is set up in between the two control boards. On the left is the servo PWM generator and on the right is the drive motor controller. The component hanging off of the servo PWM generator is the potentiometer used to angle the swerve module and the components hanging off of the drive motor controller are the switches used to operate the drive motor. When tested all functions were operational and the module could drive and angle simultaneously.

**Fig 21:** Final Swerve Module



# Conclusions

At this point in the project, I am happy to call it complete. I have successfully created a functional swerve module with angle and drive control and have learned a good amount about how low speed digital signals are handled.

My biggest takeaways are as follows: Manufacturing takes time. A large amount of delays were due to me underestimating the amount of time it would take to manufacture something, or walking into small errors which take quite a while to debug. Allocating more time to manufacturing and accounting for issues that will arise will give me a much more accurate project timeline.

I also learned that I should be sticking with mass manufactured ICs for simple components rather than making my own worse versions of them. A major setback in my project was due to me trying to create my own voltage regulator and having issues with the load on that. I solved it by moving to an LDO IC which solved all the problems I was having. Similarly, when making the comparator for my tracking ADC, just using an op amp led to problems which were solved by using an LM311. There is a reason these chips are popular: people have devoted thousands of man hours and millions of dollars into ensuring that they work as intended and it is important to rely on them.

This lab also got me much more familiar with the lab equipment and how to use it to verify circuits. And the basic underpinnings of how many common circuits work.

**Future Goals**

If I wish to continue the project, the next logical step would be to design a pcb with 4 sets of these circuits so that I can create a full swerve robot, and redesigning the modules themselves to be more compact and efficient. With this, I can create an actual swerve robot, and depending on how I lay out the PCB, I can leave expansion points available so I can use an MCU to control it or attach a radio module for remote control or any number of other advancements.